Autonomous Power Routing in Stochastic Smart Grids

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23.62% imbalance removed, 240 unseen hours

96.9% of oracle ceiling recovered

−21.2% forecast RMSE vs. persistence

0 safety or thermal violations (CPO)

ABSTRACT

Renewables and sudden demand make grid balancing hard. We compare two learning-based **power-routing controllers** on one physics-resolved stochastic simulator: a **centralized Transformer-DQN** and a **decentralized graph-based multi-agent controller**. The Transformer-DQN removes **23.62 %** of imbalance over 240 unseen hours; the decentralized controller is verified as an implementation only (PPO/CPO evaluation incomplete).

INTRODUCTION

A locally sensible transfer can overload a line elsewhere, so a controller must decide **where power goes, how much, and at what loss and risk** [1]. Research questions:

- **RQ1** How much imbalance can a Transformer-DQN remove vs. an upper bound?
- **RQ2** How does a graph-based decentralized controller behave on a shared grid?
- **RQ3** Can a safety cost guard decentralized learning?
- **RQ4** Can it transfer to a larger topology unchanged?

STOCHASTIC SIMULATOR

- Diurnal load & generation, 3-state **Markov weather**, rare events
- **DC power flow**: impedance, I²R loss, thermal limits
- Slew-rate actuators, **N−1 contingencies**

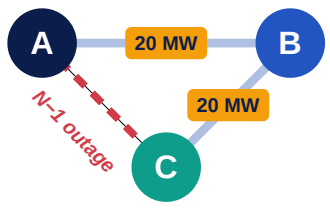


Fig. 1: Three-node grid. Imbalance at each node = G(t) − L(t).

EXPERIMENTAL SETUP

- Test: **240 unseen hours**, unmanaged baseline 16,303.73 MW
- **Greedy oracle**: best single 20 MW transfer per hour (ceiling)
- **TransferAgent**: frozen LSTM [8] encoder
- Baselines: always-Hold and random

METHODOLOGY

① CENTRALIZED · Transformer-DQN

6-hour window of 3-node imbalance

▼

Transformer encoder [4]
4 heads · d_key = 32 · positional embedding

▼

DQN Q-network [2]
replay buffer + target network

▼

7 actions: Hold + six 20 MW transfers

$$S(t) = \Sigma_i |G_i(t) - L_i(t)|$$

Reward = (imbalance removed − c_{loss}·I²R loss) / 10

② DECENTRALIZED · Graph MARL

Local observation + live adjacency matrix

▼

Graph neural network [5]
one agent per substation

▼

PPO [6] vs. safety-constrained CPO [7]

▼

Continuous ±50 MW set-points
15 MW slew limit · 0.5 actuator response

RESULTS · FORECAST

The LSTM finds real temporal structure (R^2 = 0.911), justifying a 6-hour window.

Persistence	9.57 MW
LSTM	7.54 MW

Fig. 2: Held-out RMSE (lower is better)

RESULTS · DECISIONS

59.8% oracle agreement

0.488 Cohen's κ

100% physics match

Hold is only 39.1 % of oracle labels, so agreement is not just “doing nothing”.

ACKNOWLEDGEMENT

We thank Dr. Aprna Tripathi for their guidance on this project-based learning work at Manipal University Jaipur.

RESULTS · CONTROL

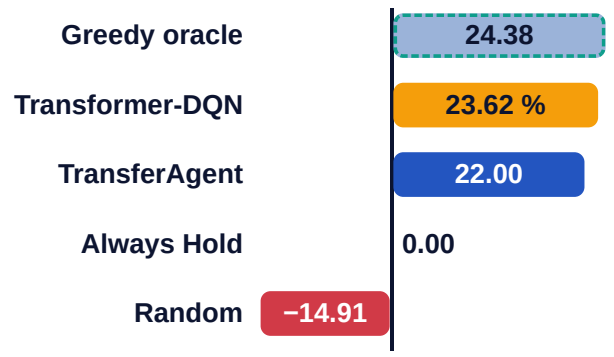


Fig. 3: Imbalance reduction (%)

Trade-off: more switching and higher I²R loss than the oracle.

DECENTRALIZED CHECK

48 h reference	Stress ↓	I²R	Viol.
No control	0.00 %	0.00	none
Proportional	**21.24 %**	**1.32 MW**	**0 %**

5 CPO seeds × 150 episodes: **zero safety cost**, no thermal/voltage violations. PPO/CPO comparison **not completed**.

CONCLUSION

1. Transformer-DQN removes **23.62 %** of imbalance: **96.9 %** of the one-step ceiling.
2. Not free: imbalance reduction and routing efficiency are distinct objectives.
3. Graph MARL has a verified foundation (observations, actuators, safety cost, topology).
4. Limits: synthetic data. Next: PPO vs. CPO, double-Q [3], larger grid.

REFERENCES

[1] J. Duan et al., “Deep-RL-based autonomous voltage control,” *IEEE Trans. Power Syst.*, 2020.

[2] V. Mnih et al., “Human-level control through deep RL,” *Nature*, 2015.

[3] H. van Hasselt et al., “Deep RL with double Q-learning,” *AAAI*, 2016.

[4] A. Vaswani et al., “Attention is all you need,” *NeurIPS*, 2017.

[5] T. Kipf and M. Welling, “Semi-supervised classification with GCNs,” *ICLR*, 2017.

[6] J. Schulman et al., “Proximal policy optimization algorithms,” arXiv:1707.06347.

[7] J. Achiam et al., “Constrained policy optimization,” *ICML*, 2017.

[8] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Comput.*, 1997.